

ECx00M-CN&ECx00N-CN&EG810M-CN

Audio Application Note

LTE Standard Module Series

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About the Document

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1 Introduction

This document explains audio related AT commands on Quectel LTE Standard ECx00M-CN, ECx00N-CN and EG810M-CN modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Series	Module
ECx00M	EC200M-CN
	EC600M-CN
	EC800M-CN
ECx00N	EC200N-CN
	EC800N-CN
-	EG810M-CN

NOTE

Due to the constraint of the module Flash storage space, different models vary in supporting audio function. For more details, please contact Quectel Technical Support.

1.2. AT Command Introduction

1.2.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

1.2.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 2: Type of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of a corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

1.3. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2 AT Commands Description

2.1. AT+CLVL Loudspeaker Volume Level Selection

This command selects the volume level of the internal loudspeaker of MT.

AT+CLVL Loudspeaker Volume Level Selection	
Test Command AT+CLVL=?	Response +CLVL: (list of supported <level>s) OK
Read Command AT+CLVL?	Response +CLVL: <level> OK Or ERROR
Write Command AT+CLVL=<level>	Response OK Or ERROR If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configuration is saved automatically.
Reference 3GPP TS 27.007	

Parameter

<level>	Integer type. Loudspeaker volume level with manufacturer specific range (Smallest value represents the lowest sound level). Range: 0–5. Default value: 3. Unit: dB.
<err>	Error codes. See Chapter 3 for more information.

2.2. AT+QAUDSW Select Codec

This command selects the built-in codec or the external one.

AT+QAUDSW Select Codec	
Test Command AT+QAUDSW=?	Response +QAUDSW: (list of supported<n>s) OK
Read Command AT+QAUDSW?	Response +QAUDSW: <n> OK
Write Command AT+QAUDSW=<n>	Response OK or ERROR If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configuration is saved automatically.
Reference 3GPP TS 27.007	

Parameter

<n>	Integer type. 0 External codec 1 Built-in codec
<err>	Error codes. See Chapter 3 for more information.

NOTE

The selection of default codec depends on PMIC chip. External codec is selected by default for PM802, and the built-in codec is selected by default for PM813.

2.3. AT+QAUDLOOP Enable/Disable Audio Loop Test

This command enables/disables audio loop test.

AT+QAUDLOOP Enable/Disable Audio Loop Test	
Test Command AT+QAUDLOOP=?	Response +QAUDLOOP: (list of supported <enable>s) OK
Read Command AT+QAUDLOOP?	Response +QAUDLOOP: <enable> OK
Write Command AT+QAUDLOOP=<enable>	Response OK Or ERROR
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configuration is not saved.

Parameter

<enable>	Integer type. Enable/disable audio loop test. 0 Disable audio loop test 1 Enable audio loop test
----------	--

2.4. AT+VTS DTMF and Tone Generation

This command sends ASCII characters that cause MSC to transmit DTMF tones to a remote subscriber. This command can only be executed during a voice call.

AT+VTS DTMF and Tone Generation	
Test Command AT+VTS=?	Response +VTS: (list of supported <DTMF>s),(list of supported of <duration>s) OK
Write Command	Response

AT+VTS=<DTMF>[,<duration>]	OK Or ERROR If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	Depends on the length of <DTMF> and <duration>
Characteristics	/
Reference 3GPP TS 27.007	

Parameter

<DTMF>	String type. ASCII characters in the set 0-9, #, *, A, B, C, D, E, F. The string should be enclosed in quotation marks ("...").
<duration>	Integer type. Duration of each tone in 1/10 seconds with tolerance. Range: 100–1000. Default value: 300. Unit: 10 ms.
<err>	Error codes. See Chapter 3 for more information.

NOTE

If the <duration> is less than the minimum time specified by the network, the actual duration will be the network specified time. If this parameter is omitted, <duration> is specified with **AT+VTD**.

Example

```

ATD12345678900;           //Dial
OK
AT+VTS="1"                 //The remote caller can hear the DTMF tone
OK
AT+VTS="1234567890A"       //Send multiple tones at a time
OK

```

2.5. AT+QAUDMOD Set Audio Mode

This command sets the audio mode required for the connected device.

AT+QAUDMOD Set Audio Mode

Test Command	Response
--------------	----------

AT+QAUDMOD=?	+QAUDMOD: (list of supported <mode>s) OK
Read command AT+QAUDMOD?	Response +QAUDMOD: <mode> OK
Write Command AT+QAUDMOD=<mode>	Response OK or ERROR If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configuration is saved automatically.
Reference Quectel	

Parameter

<mode>	Integer type. Current audio mode. 0 Echo canceller, noise suppressor, digital gain, and calibration parameter for handset 1 Echo canceller, noise suppressor, digital gain, and calibration parameter for headset 2 Echo canceller, noise suppressor, digital gain, and calibration parameter for speaker
<err>	Error codes. See Chapter 3 for more information.

2.6. AT+QDAI Digital Audio Interface Configuration

This command configures the digital audio interface.

AT+QDAI Digital Audio Interface Configuration	
Test Command AT+QDAI=?	Response +QDAI: (list of supported <io>s),(list of supported <mode>s),(list of supported <fsync>s),(list of supported <bitclock>s),(list of supported <format>s),(list of supported <sample>s),(list of supported <num_slots>s),(list of supported <slot_mapping>s)

	OK
Read Command AT+QDAI?	Response +QDAI: <io>[,<mode>,<fsync>,<bitclock>,<format>,<sample>,<num_slots>,<slot_mapping>] OK
Write Command AT+QDAI=<io>[,<mode>,<fsync>,<bitclock>[,<format>[,<sample>[,<num_slots>,<slot_mapping>]]]]	Response OK Or ERROR
Maximum Response Time	300 ms
Characteristics	This command takes effect after the module is rebooted. The configurations are saved automatically.

Parameter

<io>	Integer type. 1 Digital PCM output. You can define the PCM formats. When setting master mode, short-synchronization, 2048 kHz clock frequency, 16-bit liner data format, and 8 kHz sampling rate, the module can be used directly with default settings. 2 Analog output. External codec chip linked with PCM interface: NAU8810 model. It can be configured through the IIC. 3 Analog output. External codec chip linked with PCM interface: ALC5616 model. It can be configured through the IIC. 5 Analog output. External codec chip linked with PCM interface: TLV320AIC3104 model. It can be configured through the IIC (Currently not supported).
<mode>	Integer type. 0 Master mode 1 Slave mode (Currently not supported)
<fsync>	Integer type. 0 Primary mode (short synchronization) 1 Secondary mode (long synchronization) (Currently not supported)
<bitclock>	Integer type. Bit clock. 0 Currently not supported 1 32 FS 2 64 FS 3 128 FS 4 256 FS 5 512 FS
<format>	Integer type. Data format. 0 16-bit linear

	1	8-bit a-law (Currently not supported)
	2	8-bit u-law (Currently not supported)
<sample>		Integer type.
	0	8 kHz
	1	16 kHz
<num_slots>		Integer type.
	1	Number of slots
<slot_mapping>		Integer type. Slot mapping value. Range: 1–16 (Currently not supported).

NOTE

1. When a recommended codec is selected and 16 kHz sampling rate is required, please input the value of <sample>. Currently only ALC5616 supports 16 kHz (**AT+QDAI=3,0,0,5,0,1,1,1**).
2. Clock frequency = <bitclock>*<sample>.

Example

```

AT+QDAI=?           //Test command.
+QDAI: (1-3,5),(0,1),(0,1),(0-5),(0-2),(0,1),(1),(1-16)

OK
AT+QDAI?           //Query the current interface configuration.
+QDAI: 3,0,0,4,0,0,1,1

OK
AT+QDAI=1,0,0,4,0,0,1,1 //Set the audio output of digital PCM as the main mode, short
                        //synchronization, 8 kHz sampling rate, 2048 kHz clock frequency, 16-
                        //bit sampling bit depth, and the number of time gaps used as 1, with
                        //the first-time gap used.

OK

```

2.7. AT+QSIDET Set Side Tone Gain in Current Mode

This command sets the side tone gain value in the current mode.

AT+QSIDET Set Side Tone Gain in Current Mode

Test Command AT+QSIDET=?	Response +QSIDET: (list of supported <st_gain>s) OK
Read Command AT+QSIDET?	Response +QSIDET: <st_gain>

	OK
Write Command AT+QSIDET=<st_gain>	Response OK Or ERROR
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configuration is not saved.
Reference Quectel	

Parameter

<st_gain>	Integer type. Indicates the configured side tone gain in current mode. Range: 0–65535. Default value might be different in different audio modes.
------------------------	--

2.8. AT+QMIC Set Uplink Gains of MIC

This command sets the uplink gains of the microphone.

AT+QMIC Set Uplink Gains of MIC	
Test Command AT+QMIC=?	Response +QMIC: (list of supported <txgain>s),(list of supported <txdgain>s) OK
Read Command AT+QMIC?	Response +QMIC: <txgain>,<txdgain> OK
Write Command AT+QMIC=<txgain>[,<txdgain>]	Response OK Or ERROR
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configurations are saved automatically.

Parameter

<txgain>	Integer type. Uplink codec gain. Range: 0–65535. Default value might be different in different audio modes.
<txdgain>	Integer type. Uplink digital gain. Range: 0–65535. Default value might be different in different audio modes.

2.9. AT+QIIC Read/Write Codec via IIC

This command reads/writes codec via IIC interface.

AT+QIIC Read/Write Codec via IIC	
Test Command AT+QIIC=?	Response +QIIC: (list of supported <rw>s),(list of supported <device>s),(list of supported <addr>s),(list of supported <bytes>s),(list of supported <value>s) OK
Write Command AT+QIIC=<rw>,<device>,<addr>,<bytes>[,<value>]	Response If <rw>=0, all configuration parameters should be specified: OK If <rw>=1, <value> should be omitted: +QIIC: <value> OK
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configurations are not saved.

Parameter

<rw>	Integer type. 0 Write command 1 Read command
<device>	Hex integer type. Hexadecimal with prefix "0x". 0-0xFF 7-bit device address. (Currently supported: NAU8810: 0x1A; ALC5616: 0x1B)
<addr>	Hex integer type. Hexadecimal with prefix "0x". 0-0xFF Register address
<bytes>	Integer type. Hexadecimal with prefix "0x". 1 Write 1 byte

	2	Write 2 bytes
<value>	Hex integer type.	
	0-0xFFFF	Data value

NOTE

<device> is the address of the slave device (7-bit, does not include the read/write bit), please find it on the device datasheet.

Example

```
AT+QIIC=0,0x1B,0x00,2,0x0000 //Write 2-byte register content of the register's location: slave
                                address: 0x1B, register address:0x00, data value: 0x0000
OK
AT+QIIC=1,0x1B,0x00,2 //Read 2-byte register content of the register's location: slave
                                address: 0x1B, register address:0x00
+QIIC: 0x21
OK
```

2.10. AT+QAUDRD Record Media File

This command records the uplink or downlink speech during a voice call or records sound from local microphone in idle state and saves it to a file.

AT+QAUDRD Record Media File

Test Command AT+QAUDRD=?	Response +QAUDRD: (list of supported of <state>s),<filename>,(list of supported <format>),(list of supported <dlink>s) OK
Read Command AT+QAUDRD?	Response +QAUDRD: <state> OK
Write Command AT+QAUDRD=<control>[,<filename>[,<format>[,<dlink>]]]	Response OK Or ERROR

	If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	/

Parameter

<state>	Integer type. 0 Module is not recording sound 1 Module is recording sound
<control>	Integer type. 0 Stop recording 1 Start recording
<filename>	String type. Name of the recorded media file.
<format>	Integer type. File format. 13 WAV_PCM16
<dlink>	Integer type. Record the uplink or downlink sound*. 0 Record uplink sound 1 Record downlink sound
<err>	Error codes. See Chapter 3 for more information.

NOTE

1. **<filename>** is the path used to save the recorded file. Default path: `/data/ufs`.
2. The module supports recording format of WAV_PCM16, suffix `.wav` audio files.
3. If the name and format of the recorded file are the same as that of an existing file or an unknown error occurs, the module reports **+QAUDRIND: 0,1**.
4. If current recording is interrupted by other audio tasks, the module reports **+QAUDRIND: 0,6**.
5. If there is no space available for recording, the module reports **+QAUDRIND: 0,3**.
6. The module records uplink and downlink audio data asynchronously.
7. This command returns an error if the file format is inconsistent with file extension.

Table 3: The Description of <code> in URC +QAUDRIND: 0,<code>

<code>	Meaning
0	Reserved
1	Unknown error
3	No space for recording

6 Interrupted by other audio tasks

Example

```
AT+QAUDRD=1,"A.wav",13,0 //Record the uplink sound in WAV format and store it in UFS
OK
AT+QAUDRD=0 //Stop recording
OK
```

2.11. AT+QAUDPLAY Play Media File

This command plays local media file.

AT+QAUDPLAY Play Media File	
Test Command AT+QAUDPLAY=?	Response +QAUDPLAY: <file_name>,(list of supported <state>s) OK
Read Command AT+QAUDPLAY?	Response +QAUDPLAY: <state> OK
Write Command AT+QAUDPLAY=<file_name>,<repeat>	Response OK Or ERROR If there is any error related to ME functionality: +CME ERROR: <err> When the playback is completed: +QAUDPLAY: 0
Maximum Response Time	300 ms
Characteristics	/

Parameter

<state>	Integer type. Module status.
0	Module is not playing media file

	1	Module is playing media file
<file_name>	String type. Name of local media file, includes file path, file name and file suffix. File path must be UFS. Default file path: /data/ufs.	
<repeat>	Integer type. Play the file once or repeatedly.	
	0	Play only once
	1	Repeat
<err>	Error codes. See Chapter 3 for more information.	

NOTE

1. If an unknown error occurs, the module reports **+QAUDPIND: 0,1**.
2. If current playing is interrupted by other audio tasks, the module reports **+QAUDPIND: 0,6**.
3. This command only supports playback of audio in WAV, MP3 and ARM formats. If the audio file in the corresponding format fails to be played, **+CME ERROR: 902** will be reported.
4. The playback of media file is terminated if the ringing tone is on.
5. If the <file_name> and <repeat> are set to invalid values, the module reports **+CME ERROR: 902**; if the local media file does not exist, the module reports **+CME ERROR: 903**.
6. For more information about <code> in URC **+QAUDRIND: 0,<code>**, see **Table 3**.

Example

```
AT+QAUDPLAY="A.mp3",0      //Play an mp3 file stored in UFS only once
OK
```

```
+QAUDPLAY: 0
AT+QAUDPLAY="A.mp3",1      //Play an mp3 file stored in UFS repeatedly
OK
```

```
+QAUDPLAY: 0                //Playback is completed
```

2.12. AT+QAUDPLAYSPL Play Spliced Media File

This command plays spliced media file.

AT+QAUDPLAY Play Spliced Media File

Test Command	Response
AT+QAUDPLAYSPL=?	+QAUDPLAYSPL: <filenameX>
	OK
Read Command	Response
AT+QAUDPLAYSPL?	+QAUDPLAYSPL: <state>

	OK
Write Command AT+QAUDPLAYSPL=<filename1>,<filename2>, ...	Response OK Or ERROR If there is any error related to ME functionality: +CME ERROR: <err> When the playback is completed: +QAUDPLAYSPL: 0
Maximum Response Time	300 ms
Characteristics	/

Parameter

<state>	Integer type. Module status. 0 Module is not playing media file 1 Module is playing media file
<filenameX>	String type. Full path to the local media file, which consists of the audio file path, file name and file suffix. Default file path: <i>/data/ufs</i> . Only the audio file in WAV format is supported, and at least two WAV format files should be set.
<err>	Error codes. See Chapter 3 for more information.

NOTE

1. When the module is not powered on completely, the audio playback will be abnormal. It is recommended to wait at least 20 seconds after receiving **RDY** to ensure that the network registration is normal before playing audio files.
2. This command is only supported by EC200N-CN and EC800N-CN modules.

2.13. AT+QAUDSTOP Stop Playing Media File

This command stops playing media file.

AT+QAUDSTOP Stop Playing Media File

Test Command AT+QAUDSTOP=?	Response OK
--------------------------------------	-----------------------

Execution Command AT+QAUDSTOP	Response OK If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	/

Parameter

<err>	Error codes. See Chapter 3 for more information.
--------------------	---

2.14. AT+QTONEDT Enable/Disable DTMF Detection

This command enables or disables DTMF detection. If this function is enabled, DTMF tones sent by the other side will be detected and reported on the assigned serial port.

AT+QTONEDT Enable/Disable DTMF Detection	
Test Command AT+QTONEDT=?	Response +QTONEDT: (list of supported <enable>s) OK
Read Command AT+QTONEDT?	Response +QTONEDT: <enable> OK
Write Command AT+QTONEDT=<enable>	Response OK Or ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<enable>	Integer type. Enable or disable DTMF detection. <u>0</u> Disable
-----------------------	---

1 Enable

NOTE

DTMF characters - ASCII:

DTMF	ASCII	DTMF	ASCII
0	48	8	56
1	49	9	57
2	50	A	65
3	51	B	66
4	52	C	67
5	53	D	68
6	54	*	42
7	55	#	35
E	69	F	70

2.15. AT+QAUDUPDATE Update Audio Debugging Parameters

This command updates the audio debugging parameters.

AT+QAUDUPDATE Update Audio Debugging Parameters

Test Command AT+QAUDUPDATE=?	Response +QAUDUPDATE: (list of supported <func>s),(list of supported <file>s) OK
Read Command AT+QAUDUPDATE?	Response OK
Write Command AT+QAUDUPDATE=<func>[,<file>]	Response OK or ERROR If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	/

Parameter

<func>	Integer type. 0 Update parameters 1 Reserved
<file>	Integer type. Name of audio debugging parameter file. 0 audio_config.nvm 1 audio_gain.nvm 2 audio_ve.nvm

NOTE

1. You should upload audio debugging parameter files with **AT+QFUPL** to the file system before updating the parameters of the files. For more information about **AT+QFUPL**, see **document** *错误! 未找到引用源。*.
2. For more information about obtaining audio debugging parameter files, contact Quectel Technical Support.

Example

```
AT+QFUPL="audio_config.nvm",298 //Upload 298 B calibration file
OK
AT+QAUUPDATE=0,0 //Update parameters for audio_config.nvm
OK
```

2.16. AT+QPSND Play Audio File

This command plays a local audio file.

AT+QPSND Play Audio File

Test Command AT+QPSND=?	Response +QPSND: (list of supported <control>s), <file_name> ,(list of supported <repeat>s),(list of supported <ulmute>s),(list of supported <dlmute>s) OK
Read Command AT+QPSND?	Response +QPSND: <state> OK

Write Command AT+QPSND=<control>,<file_name>,<repeat>[,<ulmute>[,<dlmute>]]	Response OK Or ERROR If there is any error related to ME functionality: +CME ERROR: <err> When the playback is completed: +QPSND: 0
Maximum Response Time	300 ms
Characteristics	/

Parameter

<control>	Integer type. 0 Stop playing 1 Start playing
<file_name>	String type. Name of local audio file, including file path, file name and file suffix. Default playback path: / data/ufs.
<repeat>	Integer type. Play the file once or repeatedly. 0 Play only once 1 Repeat
<ulmute>	Integer type. Mute or not mute uplink. 0 Mute 1 Do not mute
<dlmute>	Integer type. Mute or not mute downlink. 0 Mute 1 Do not mute
<state>	Integer type. 0 Module is not playing the file 1 Module is playing the file
<err>	Error codes. See Chapter 3 for more information.

NOTE

The module only supports mono WAV and MP3 formats.

Example

```
AT+QPSND=1,"A.wav",0 //Play a file in WAV format, and the file is stored in UFS.
```

OK

+QPSND: 0

AT+QPSND=1,"A.wav",0,1,1

//Play a file in WAV format to far-end once when a call is ongoing

OK

+QPSND: 0

2.17. AT+QLDTMF Play Local DTMF

This command plays a local DTMF string. The maximum length of a local DTMF string is 20 characters. It can also be used to stop playing the DTMF string.

AT+QLDTMF Play Local DTMF

Test Command AT+QLDTMF=?	Response +QLDTMF: (list of supported <n>s),(list of supported <DTMF>s) OK
Write Command AT+QLDTMF=<n>,<DTMF>[,<y>]	Response OK Or ERROR If there is any error related to ME functionality: +CME ERROR: <err> When the playback is completed: +QLDTMF: 5
Execute Command AT+QLDTMF	Response OK
Maximum Response Time	300 ms
Characteristics	/

Parameter

<n>	Integer type. Play time and mute time of each DTMF. Range: 1–1000. Unit: 1/100 second when <y>=1, or 1/10 second when <y> is not set.
<DTMF>	String type. DTMF string. Maximum length: 20 characters, separated by comma.

<y>	DTMF format: 0-9, *, #, A-F. The string should be enclosed in quotation marks ("..."). Integer type. If this parameter is omitted, means the unit of <n> is 1/10 second. If this parameter is specified to 1, it means the unit of <n> is 1/100 second.
<err>	Error codes. See Chapter 3 for more information.

Example

```

AT+QLDTMF=? //Test command
+QLDTMF: (1-1000),(0-9,*,#,A-F)

OK
AT+QLDTMF=2,"A,B,1,2,#" //Play "A,B,1,2,#", play time & mute time are both 200 ms

OK
AT+QLDTMF //Stop playing
OK

```

2.18. AT+QWDTMF Play/Send DTMF Tones to Far-end

This command plays or sends DTMF tones to far-end.

AT+QWDTMF Play/Send DTMF Tones to Far-end	
Test Command AT+QWDTMF=?	Response +QWDTMF: (list of supported <ulmute>s),(list of supported <dlmute>s),(list of supported <DTMF>s),(list of supported <duration>s) OK
Write Command AT+QWDTMF=<ulmute>,<dlmute>,<DTMF>,<duration>	Response OK or ERROR If there is any error related to ME functionality: +CME ERROR: <err> When the playback is completed: +QWDTMF: 5
Write Command AT+QWDTMF=<ulmute>,<dlmute>,<value>	Response OK or

	ERROR If there is any error related to ME functionality: +CME ERROR: <err> When the playback is completed: +QWDTMF: 5
Execution Command AT+QWDTMF	Response If the optional parameters are omitted, stop playing DTMF: OK
Maximum Response Time	300 ms
Characteristics	/

Parameter

<ulmute>	Integer type. Mute or not mute uplink. 0 Mute 1–7 Do not mute (Volume level setting is not currently supported.)
<dlmute>	Integer type. Mute or not mute downlink. 0 Mute 1–7 Do not mute (Volume level setting is not currently supported.)
<DTMF>	String type. Maximum length: 64 characters, separated by comma. DTMF format: 0–9, *, #, A–F. The string should be enclosed in quotation marks ("...").
<duration>	The duration of play time and mute time. Range: 500-1000. Unit: ms.
<value>	String type. Sets of DTMF tone, play time and mute time, separated by comma. DTMF tone Range: 0–9, *, #, A–F. No comma needed between two tones. Maximum length: 64 characters. Play time Range: 100-1000. Unit: ms. Mute time Range: 100-1000. Unit: ms. For example: "AB,500,500,DCD,100,1000" means AB DTMF tones, 500 ms play time, 500 ms mute time; DCD DTMF tones, 100 ms play time, 1000 ms mute time.

Example

AT+QWDTMF=?	//Test command
+QWDTMF: (0-7),(0-7),(0-9,*,#,A-F),(500-1000)	
OK	
AT+QWDTMF=1,0,"AB,1000,500,DCD,100,1000"	//Play AB DTMF, with play time 1000 ms and mute time 500 ms; Play DCD DTMF, with play time 100 ms and mute time 1000 ms
OK	

```

AT+QWDTMF                                     //Stop playing
OK
AT+QWDTMF=1,0,"1,2,3,A,B",500                //Play 123AB DTMF, with play time 500 ms,
                                                and mute time 500 ms
OK

```

NOTE

Some modules do not support E and F DTMF tones because it is a single tone. If set, an error will be reported. Please contact Quectel Technical Support for details.

2.19. AT+QLTONE Play Local Customized Tones

This command plays a local customized tone.

AT+QLTONE Play Local Customized Tones

Test Command AT+QLTONE=?	Response +QLTONE: (list of supported <mode>s),(list of supported <frequency>s),(list of supported <period_on>s),(list of supported <period_off>s),(list of supported <duration>s) OK
Write Command AT+QLTONE=<mode>[,<frequency>,<period_on>,<period_off>,<duration>]	Response OK Or ERROR If there is any error related to ME functionality: +CME ERROR: <err> When the playback is completed: +QLTONE: 0
Execution Command AT+QLTONE	Response OK Or ERROR
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configurations are not saved.

Parameter

<mode>	Integer type. 0 Stop playing 1 Start playing
<frequency>	Integer type. Tone frequency. Range: 100–3500. Unit: Hz.
<period_on>	Integer type. Tone play time. Range: 0–1000. Unit: ms.
<period_off>	Integer type. Tone mute time. Range: 0–1000. Unit: ms.
<duration>	Integer type. Tone total time. Range: 0–15300000. Unit: ms.
<err>	Error codes. See Chapter 3 for more information.

Example

```

AT+QLTONE=? //Test command
+QLTONE: (0,1),(100-4000),(0-1000),(0-1000),(0-15300000)

OK
AT+QLTONE=1,1000,200,300,3000 //Play a 1000 Hz tone, with play time of 200 ms and mute time
                                300 ms. Total time: 3000 ms

OK

+QLTONE: 0
AT+QLTONE=0 //Stop playing
OK
AT+QLTONE //Stopped playing
OK

```

2.20. AT+QAUDHSDDET Set Headset Detection and Earphone Function

This command enables or disables headphone detection function while turning on or turning off the earphone.

AT+QAUDHSDDET Set Headset Detection and Earphone Function

Test Command	Response
AT+QAUDHSDDET=?	+QAUDHSDDET: (list of supported <enable>s),(list of supported <type>s)
	OK

Read Command AT+QAUDHSDDET?	Response +QAUDHSDDET: <enable>,<type>
	OK
Write Command AT+QAUDHSDDET=<enable>,<type>	Response OK Or ERROR
	If there is any error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	This command takes effect after the module is rebooted. The configurations will be saved automatically.

Parameter

<enable>	Integer type. Enable or disable the headset detection function. 0 Disable 1 Enable
<type>	Integer type. 0 Normally open earphone 1 Normally close earphone
<err>	Error codes. See Chapter 3 for more information.

NOTE

1. This command is only applicable to the built-in codec, therefore switch to built-in codec before executing this command.
2. The module that does not support headphone function does not support this command. For details, contact Quectel Technical Support.

2.21. AT+QDTMFVOL Set Local Tone Volume

This command sets the volume of local tone.

AT+QDTMFVOL Set Local Tone Volume

Test Command AT+QDTMFVOL=?	Response +QDTMFVOL: (list of supported <level>s)
--------------------------------------	--

	OK
Read Command AT+QDTMFVOL?	Response +QDTMFVOL: <level>
	OK
Write Command AT+ QDTMFVOL=<level>	Response OK or ERROR
Maximum Response Time	300 ms
Characteristics	This command takes effect immediately. The configuration is not saved.

Parameter

<level> Integer type. Volume level. Range: 0–10 (Smallest value represents the lowest sound level).

2.22. AT+QCFG="tone/incoming" Configure Incoming Call Ringtone

This command configures the ringtone for incoming calls.

AT+QCFG="tone/incoming" Configure Incoming Call Ringtone

Test Command AT+QCFG=?	Response ... +QCFG: "tone/incoming", (list of supported <enable>s) ... OK
Write Command AT+QCFG="tone/incoming" [, <enable>]	Response If the optional parameter is omitted, query the current configuration: +QCFG: "tone/incoming", <enable> OK If the optional parameter is specified, configure the ringtone for incoming calls: OK Or

	ERROR
Maximum Response Time	150 ms
Characteristics	This command takes effect immediately; The configuration is saved automatically.

Parameter

<enable>	Integer type. Incoming call ringtone configuration.
<u>0</u>	Turn off ringtone
1	Turn on the ringtone, with the type of customized ringtone. You need to rename the ringtone to <i>Tone.mp3</i> and upload it to the UFS directory of the module's file system.
2	Turn on ringtone, and the ringtone is a “beep” sound, with a frequency of around 450Hz sinusoidal waveform

Example

```

AT+QCFG="tone/incoming",2    //Turn on ringtone
OK
AT+QCFG="tone/incoming"      //Query the current configuration
+QCFG: "tone/incoming",2
OK

```

3 Error Codes

Table 4: Error Codes of +CME ERROR: <err>

Code of <err>	Meaning
901	Audio unknown error
902	Audio invalid parameters
903	Audio operation not supported
904	Audio device busy

4 Appendix References

Table 5: Related Document

Document Name
[1] Quectel_ECx00M&ECx00N&EC800K_Series_FILE_Application_Note

Table 6: Terms and Abbreviations

Abbreviation	Description
ASCII	American Standard Code for Information Interchange
DCD	Data Carrier Detection
DTMF	Dual Tone Multiple Frequency
GBK	Chinese Internal Code Specification
IIC	Inter-Integrated Circuit
LTE	Long-Term Evolution
ME	Mobile Equipment
MIC	Microphone
MSC	Mobile Switching Center
NVM	Non-Volatile Memory
PCM	Pulse Code Modulation
PMIC	Power Management Integrated Circuit
TA	Terminal Adapter
UCS	Universal Character Set
UFS	User File System

URCUnsolicited Result Code
